

NAMPO, Korea, Democratic People's Republic of

WMO: 470600

Lat: 38.717N

Lon: 125.383E

Elev: 154

StdP: 14.61

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	8.8	12.2	-3.1	4.7	11.5	0.6	5.7	14.7	24.3	17.5	20.7	22.1	6.8	320	0.401

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	10.9	87.7	79.2	85.3	77.4	83.2	76.0	80.6	86.1	79.1	83.7	77.8	81.7	4.3	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
79.0	151.3	84.3	77.7	144.7	82.4	76.6	139.3	80.8	44.4	86.3	42.9	84.0	41.5	81.8	87.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
20.5	17.0	14.1	5.1	90.8	5.1	2.5	1.4	92.6	-1.6	94.1	-4.4	95.5	-8.2	97.4		
			4.0	82.7	4.5	2.4	0.8	84.4	-1.9	85.8	-4.4	87.1	-7.7	88.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	52.7	23.9	28.6	39.5	52.0	62.9	71.3	76.7	77.5	69.4	57.7	42.5	29.0
	DBStd	19.58	7.02	6.77	6.16	6.14	4.81	3.76	3.66	3.86	4.20	5.92	8.14	7.31
	HDD50	2698	810	598	331	49	0	0	0	0	12	246	652	
	HDD65	5616	1275	1018	792	391	99	4	0	13	232	675	1117	
	CDD50	3681	0	0	4	108	399	639	826	851	581	251	22	0
	CDD65	1124	0	0	0	1	33	193	361	386	144	6	0	0
	CDH74	6738	0	0	0	5	160	792	2396	2898	475	12	0	0
	CDH80	1433	0	0	0	0	17	116	536	730	34	0	0	0
	WSAvg	5.2	5.3	5.5	6.6	6.5	6.0	4.8	4.3	4.3	4.3	4.4	5.1	5.2
	PrecAvg	33.2	0.4	0.6	0.7	1.7	3.0	3.6	9.8	7.2	3.5	1.5	1.4	0.7
Precipitation	PrecMax	70.2	1.1	1.4	1.9	3.9	7.0	11.2	23.3	19.5	14.4	4.4	4.5	1.8
	PrecMin	16.1	0.1	0.0	0.0	0.1	0.2	0.8	2.3	1.0	0.4	0.1	0.2	0.1
	PrecStd	10.8	0.3	0.4	0.5	0.9	2.0	2.3	5.5	4.7	3.3	1.1	1.0	0.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	42.8	50.0	61.9	73.7	82.1	85.7	90.5	90.8	83.3	75.7	64.2	48.6
		MCWB	39.3	42.8	53.0	60.2	66.7	71.4	80.1	81.7	73.3	65.0	57.1	45.1
	2%	DB	39.2	44.5	56.0	68.9	77.9	83.0	87.4	88.4	80.8	71.9	59.9	44.8
		MCWB	35.7	39.9	48.0	57.9	64.3	71.4	79.0	80.2	71.8	63.1	55.2	41.3
	5%	DB	36.5	41.1	52.4	65.6	75.1	80.9	85.0	86.3	78.8	69.1	56.7	41.7
		MCWB	33.6	36.6	46.0	55.4	62.7	70.1	77.7	78.9	70.1	61.3	52.6	38.6
	10%	DB	33.8	38.2	48.8	62.3	72.2	78.6	82.9	84.0	76.9	66.5	53.9	39.1
		MCWB	31.3	34.7	43.0	53.5	61.2	69.2	76.7	77.4	69.0	59.8	49.7	36.2
	0.4%	WB	40.5	45.2	54.3	62.9	69.9	75.7	82.6	83.2	76.7	67.3	59.7	46.3
		MCDB	41.9	47.5	60.0	70.7	79.1	81.8	88.0	88.9	80.5	72.9	62.4	48.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	36.4	40.5	49.9	59.4	67.0	73.9	80.4	81.0	74.4	64.7	55.9	41.9
		MCDB	38.7	43.9	54.7	66.9	74.6	80.0	85.5	86.7	78.3	69.7	58.8	44.3
	5%	WB	33.9	37.4	46.6	57.0	64.9	72.3	78.8	79.8	72.4	62.9	53.3	38.9
		MCDB	36.1	40.6	51.1	63.7	71.9	77.9	83.2	84.8	76.6	67.4	56.6	41.3
	10%	WB	31.6	35.0	43.7	54.6	63.0	70.9	77.5	78.6	70.7	61.1	50.4	36.4
		MCDB	33.7	37.8	48.4	60.4	69.5	76.1	81.4	82.8	75.0	65.5	53.3	38.7
Mean Daily Temperature Range	5% DB	MDBR	11.3	12.0	13.0	14.9	15.7	13.3	10.4	10.9	13.3	14.1	11.9	10.3
		MCDBR	13.7	15.9	19.1	20.1	20.3	17.0	13.5	13.3	15.3	16.3	15.3	12.7
		MCWBR	11.6	12.3	12.9	11.7	10.1	8.4	7.2	7.0	8.2	9.2	11.6	10.9
	5% WB	MCDBR	12.8	15.0	17.9	18.0	18.0	14.0	11.3	11.9	12.6	14.8	14.5	12.5
		MCWBR	11.5	12.6	13.5	12.0	10.7	8.2	6.9	7.0	8.6	9.7	12.2	11.4
Clear-Sky Solar Irradiance	taub		0.424	0.492	0.551	0.618	0.633	0.717	0.691	0.588	0.516	0.534	0.504	0.436
	taud		1.933	1.809	1.723	1.638	1.656	1.559	1.658	1.874	1.973	1.825	1.837	1.929
	Ebn at Noon		220	221	223	218	219	201	205	224	232	206	191	204
	Edh at Noon		43	56	68	79	79	87	78	62	52	55	48	41
All-Sky Solar Radiation	RadAvg		758	1011	1390	1623	1804	1658	1330	1426	1391	1116	746	649
	RadStd		64	76	108	118	165	168	172	169	118	71	76	44

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (26 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+157	+94

Nomenclature: See separate page